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w4.1- 3.1 - Testing

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w4.1- 3.1 - Testing.mp4

Testing

Two big testing phases:

1. **Component Testing** – Testing individual components
2. **Integration Testing** – Testing components working together

So and
then once you're done integration testing,

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w4.1- 3.1 - Testing.mp4

Component Testing

Testing one component

1. **Apply test data**
2. **Observe test results**

- Test data should be "**complete**"
  - Cover all possibilities
  - May not be possible
  - Cover all possibilities which are significantly different

then the rest of them are just
basically in the same equivalent set

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w4.1- 3.1 - Testing.mp4

Component Testing

Testing one component

1. **Apply test data**
2. **Observe test results**

- Test data should be "**complete**"
  - Cover all possibilities
  - May not be possible
  - Cover all possibilities which are significantly different
- Ex. 32-bit divider
  - Try all 64-bit inputs?
  - Try positive, negative, div by zero, etc.

Here's what I mean by that.

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w4.1- 3.1 - Testing.mp4

Integration Testing

- Test groups of connected components together
- Perform integration incrementally
- Ex. Multicopter test
  1. Test flight controller
  2. Test ESCs
  3. Test flight controller + ESCs
  4. Test flight controller + ESCs + motors

I know the fly control and the ESC works separately.

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Summary

1. The Two Pillars of Testing

You cannot simply build a complex system and flip the switch. You must move through two distinct phases:

- **Component Testing:** Testing individual parts in isolation (e.g., testing just the motor or just the sensor).
- **Integration Testing:** Testing how those parts behave once they are wired together.

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2. Component Testing: Hardware vs. Software

The way you "feed" data to a component depends on its nature:

- **Software:** You call functions with different arguments or provide command-line inputs.
- **Hardware:** You physically wire inputs, such as connecting buttons or toggles, to simulate data.

The "32-Bit Divider" Rule (Interesting Inputs)

You can almost never test every possible input. For example, a 32-bit divider has 2^{64} combinations—too many to test! Instead, you must pick **Interesting Inputs**:

- **Edge Cases:** Dividing by zero.
- **Polarity:** Positive vs. negative numbers.
- **Scale:** Very large numbers vs. very small numbers.

> **Ninja Tip:** If your system is simple (only a few inputs), test **everything**. If it's complex, manually pick the inputs most likely to cause a "bug."

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3. Integration Testing: The Incremental Approach

The biggest mistake students make is the **"Big Bang" approach**—wiring everything at once and hoping it works. When it fails (and it will), you won't know where the error is.

The Incremental Strategy:

1. Test Part A.
2. Test Part B.
3. Connect A to B and test the pair.
4. Add Part C and test the trio.

Example: The Multicopter

- **Step 1:** Test the Flight Controller (Center).
- **Step 2:** Test the ESCs (Electronic Speed Controllers).
- **Step 3:** Connect Flight Controller to ESCs. If it fails here, you know the **communication** between them is the problem.
- **Step 4:** Add the Motors. If it fails now, you know the motors (or their wiring) are faulty.